USER INTERFACE DESIGN EXP:4

MEMORY CHALLENGE USING VISUAL RECALL

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Aim

To design and evaluate a Memory Challenge user interface that demonstrates visual short-term memory recall using timed memorization and recall phases.

Objective

- To understand the concept of visual memory in Human–Computer Interaction.
- To design interactive memory UI screens.
- To analyse how grid-based layout improves recall accuracy.
- To measure user performance through a result screen.

Tools Used

- Figma (UI Design & Prototyping)
- Desktop / Laptop
- Visual Layout & Interaction Design Principles

Theory

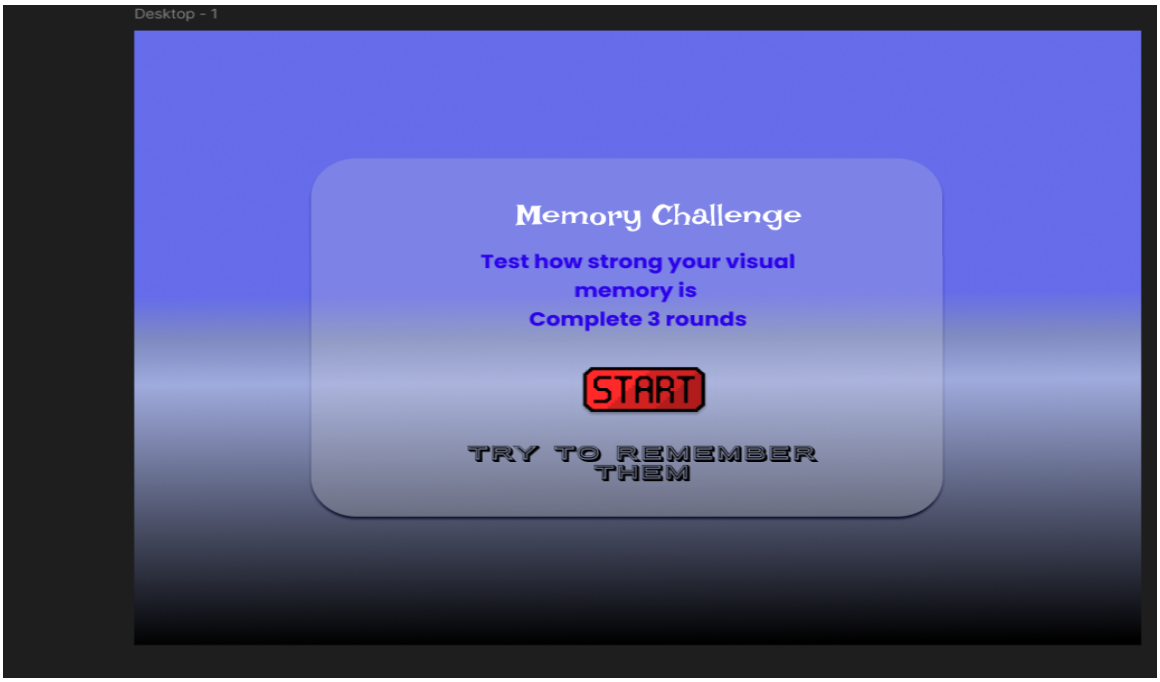
Visual memory refers to the ability of individuals to retain and recall visual information for a short duration. According to cognitive psychology, short-term memory can hold limited items at a time. When visual elements are arranged in structured layouts, cognitive load is reduced and recall efficiency increases.

In UI/UX design, structured grids, spacing, and visual grouping help users mentally organize information and improve recognition.

Experimental Design

- Home Screen (Instruction Page)
- Memory Display Phase (Memorization Round)
- Recall Phase
- Result Screen

A. Home Screen (Instruction Page)



Description

This screen introduces the Memory Challenge task. It contains a title, short instructions and a START button to begin the experiment. The centered card layout and gradient background focus user attention.

B. Memory Display Phase

Key Design Features

- Grid-based object arrangement
- Uniform spacing and alignment
- Countdown timer to limit viewing time
- Round indicator for structured gameplay
- Clear visual hierarchy

C. Recall Phase

UI Elements

- Clickable objects
- Modified grid with distractor items
- Timer indicator
- Round label
- Instruction text

D. Result Screen

Information Shown

- Score percentage (80%)
- Progress bar indicating performance
- Feedback message
- Restart and End buttons

Observation

- Structured grid layout improved recall accuracy.
- Countdown timer increased user focus.
- Progress bar enhanced engagement.
- Interactive recall encouraged active participation.

Result

The experiment shows that structured visual presentation combined with timed memorization improves short-term memory recall.

Conclusion

This experiment demonstrates the practical application of cognitive principles in UI design. By using grid structure, timed exposure, and feedback mechanisms, user memory performance and engagement were significantly improved.